

Unit - 11

Dictionary in python

```
In [2]: a = { "colour" : "red",
              "year" : 1997,
              "sports" : "cricket"}
print(a)
print(type(a))

{'colour': 'red', 'year': 1997, 'sports': 'cricket'}
<class 'dict'>
```

```
In [3]: a = { "colour" : "red",
              "year" : 1997,
              "sports" : "cricket"}
print(a["year"])
print(type(a))

1997
<class 'dict'>
```

duplicates not allowed

```
In [4]: a = { "colour" : "red",
              "year" : 1997,
              "sports" : "cricket",
              "sports" : "football",
              "sports" : "tennis"}
print(a)
print(type(a))

{'colour': 'red', 'year': 1997, 'sports': 'tennis'}
<class 'dict'>
```

```
In [5]: a = { "colour" : "red",
              "year" : 1997,
              "sports" : "cricket",
              "sports" : "football",
              "sports" : "tennis",
              "sports" : "basket ball"}
print(a)
print(type(a))

{'colour': 'red', 'year': 1997, 'sports': 'basket ball'}
<class 'dict'>
```

dictionary length

```
In [1]: d = { "apple": "fruit", "red":"colour" , "north":"direction", 2:"code"}
print(len(d))
```

4

dictionary item - data types

```
In [2]: a = { "colour":"red", 12:14, 2:56.3, "apple":[1,2,3], 3:(-1,-2,-3), "boy":{2,4}
print(a)
print(type(a))
```

```
{'colour': 'red', 12: 14, 2: 56.3, 'apple': [1, 2, 3], 3: (-1, -2, -3), 'boy': {2, 4, 6}}
<class 'dict'>
```

dict() constructor

```
In [3]: n = dict( name = "priya", age = 24, gender = "female")
print(n)
print(type(n))
```

```
{'name': 'priya', 'age': 24, 'gender': 'female'}
<class 'dict'>
```

accessing items

you can access the items of a dictionary by referring to its key name inside the square bracket

```
In [4]: n = { 1:"apple", 2:"boy", 3:"cat"}
print(n)
print(n[2])
print(n[3])
```

```
{1: 'apple', 2: 'boy', 3: 'cat'}
boy
cat
```

get() method

```
In [5]: n = { 1:"apple", 2:"boy", 3:"cat"}
x = n.get(2)
print(x)
```

boy

keys()

```
In [7]: m={"apple":12, 25:"hello", 26.3:"cat", "python":-12}
x = m.keys()

print(x)
```

dict_keys(['apple', 25, 26.3, 'python'])

```
In [8]: m={"apple":12, 25:"hello", 26.3:"cat", "python":-12}
print(m.keys())
```

dict_keys(['apple', 25, 26.3, 'python'])

values()

```
In [10]: m={"apple":12, 25:"hello", 26.3:"cat", "python":-12}
x = m.values()
print(x)
print(type(x))
```

dict_values([12, 'hello', 'cat', -12])
<class 'dict_values'>

```
In [11]: m={"apple":12, 25:"hello", 26.3:"cat", "python":-12}
print(m.values())
```

dict_values([12, 'hello', 'cat', -12])

items()

```
In [12]: m={"apple":12, 25:"hello", 26.3:"cat", "python":-12}
x = m.items()
print(m)
print(type(m))
```

{'apple': 12, 25: 'hello', 26.3: 'cat', 'python': -12}
<class 'dict'>

```
In [14]: m={"apple":12, 25:"hello", 26.3:"cat", "python":-12}
x = m.items()
print(m)
print(type(m))
```

```
{'apple': 12, 25: 'hello', 26.3: 'cat', 'python': -12}
<class 'dict'>
```

check if key exist

```
In [15]: m={"apple":12, 25:"hello", 26.3:"cat", "python":-12}
if 25 in m:
    print("yes")
```

```
yes
```

```
In [18]: m={"apple":12, 25:"hello", 26.3:"cat", "python":-12}
if "boy" in m:
    print("yes")
else:
    print("no")
```

```
no
```

change values

```
In [19]: b = {10:"hello", 11:"world", 9:[12,13,14]}
b[9] = ["a", "b", "c"]
print(b)
```

```
{10: 'hello', 11: 'world', 9: ['a', 'b', 'c']}
```

update()

```
In [20]: b = {10:"hello", 11:"world", 9:[12,13,14]}
b.update({9:["a", "b", "c"]})
print(b)
```

```
{10: 'hello', 11: 'world', 9: ['a', 'b', 'c']}
```

```
In [21]: a = {1:"apple", 2:"boy", 3:"cat", 4:"datascience"}
a.update({2:(45,46,47)})
print(a)
```

```
{1: 'apple', 2: (45, 46, 47), 3: 'cat', 4: 'datascience'}
```

adding items

```
In [22]: p = {"colour":"red", "country":"Inida", "year":[2023,2024,2025]}
p["fruit"] = "apple"
print(p)
```

```
{'colour': 'red', 'country': 'Inida', 'year': [2023, 2024, 2025], 'fruit': 'apple'}
```

```
In [24]: p = {"colour":"red", "country":"Inida", "year":[2023,2024,2025]}
p["fruit"] = "apple"
print(p)
p[9]="python is so easy"
print(p)
```

```
{'colour': 'red', 'country': 'Inida', 'year': [2023, 2024, 2025], 'fruit': 'apple'}
{'colour': 'red', 'country': 'Inida', 'year': [2023, 2024, 2025], 'fruit': 'apple', 9: 'python is so easy'}
```

update dictionary

```
In [27]: p = {"colour":"red", "country":"India", "year":[2023,2024,2025]}
p.update({"country":"Bharat"})
print(p)
p.update({"capital":"New delhi"})
print(p)
```

```
{'colour': 'red', 'country': 'Bharat', 'year': [2023, 2024, 2025]}
{'colour': 'red', 'country': 'Bharat', 'year': [2023, 2024, 2025], 'capital': 'New delhi'}
```

removing items

pop() method

```
In [28]: a = { 10:"hello", 12:"program" , 13:(1,2,3)}
a.pop(12)
print(a)
```

```
{10: 'hello', 13: (1, 2, 3)}
```

popitem() method

```
In [29]: a = { 10:"hello", 12:"program" , 13:(1,2,3)}
a.popitem()
print(a)

{10: 'hello', 12: 'program'}
```

del() method

```
In [30]: a = {"a":"hello", "b":"world", "c":(10,11,12)}
del a["b"]
print(a)

{'a': 'hello', 'c': (10, 11, 12)}
```

```
In [31]: a = {"a":"hello", "b":"world", "c":(10,11,12)}
del a
print(a)
```

```
-----
NameError                                Traceback (most recent call last)
~\AppData\Local\Temp\ipykernel_2488\1663620895.py in <module>
      1 a = {"a":"hello", "b":"world", "c":(10,11,12)}
      2 del a
----> 3 print(a)

NameError: name 'a' is not defined
```

clear() method

```
In [34]: a = {1:"hello", 2:23,3:36}
a.clear()
print(a)
print(type(a))

{}
<class 'dict'>
```

copy() method

```
In [38]: """ why you need copy method"""  
a = {1:[23,25,28], "code":(-1,-2,-3), "India":"Best country"}  
b = a  
print(b)  
a.update({"India":"best country in the world"})  
print(a)  
print(b)
```

```
{1: [23, 25, 28], 'code': (-1, -2, -3), 'India': 'Best country'}  
{1: [23, 25, 28], 'code': (-1, -2, -3), 'India': 'best country in the world'}  
{1: [23, 25, 28], 'code': (-1, -2, -3), 'India': 'best country in the world'}
```

```
In [42]: a = {1:[23,25,28], "code":(-1,-2,-3), "India":"Best country"}  
b = a.copy()  
print(b)  
b.update({"India":"best country in the world"})  
print(a)  
print(b)
```

```
{1: [23, 25, 28], 'code': (-1, -2, -3), 'India': 'Best country'}  
{1: [23, 25, 28], 'code': (-1, -2, -3), 'India': 'Best country'}  
{1: [23, 25, 28], 'code': (-1, -2, -3), 'India': 'best country in the world'}
```

make a copy of a dictionary with the dict() function

```
In [43]: a = {1:"clours", 2:"India", 3:[10,11,23]}  
b = dict(a)  
print(b)  
b.update({3:"education"})  
print(b)  
print(a)
```

```
{1: 'clours', 2: 'India', 3: [10, 11, 23]}  
{1: 'clours', 2: 'India', 3: 'education'}  
{1: 'clours', 2: 'India', 3: [10, 11, 23]}
```

Nested Dictionaries

```
In [45]: a = {
    "class1":{"name":"anshul", "section":"A", "age":24},
    "class2":{"name":"priya", "section":"B", "age": 21},
    "class3":{"name":"ayush","section":"C", "age":18}
}
print(a)
print(a["class3"]["name"])

{'class1': {'name': 'anshul', 'section': 'A', 'age': 24}, 'class2': {'name': 'priya', 'section': 'B', 'age': 21}, 'class3': {'name': 'ayush', 'section': 'C', 'age': 18}}
```

ayush

fromkeys() method

```
In [47]: x = ("key1", "key2", "key3")
y = 0
thisdict = dict.fromkeys(x,y)
print(thisdict)

{'key1': 0, 'key2': 0, 'key3': 0}
```

```
In [48]: x = ("key1", "key2", "key3")
y = 23
thisdict = dict.fromkeys(x,y)
print(thisdict)

{'key1': 23, 'key2': 23, 'key3': 23}
```

```
In [49]: x = ("key1", "key2", "key3")
a = dict.fromkeys(x)
print(a)

{'key1': None, 'key2': None, 'key3': None}
```

setdefault() Method

The setdefault() method returns the value of the item with the specified key. If the key does not exist, insert the key, with the specified value,

```
In [50]: movie = { "genre": "comedy",
    "title":"spiderman",
    "year":2000
}
x = movie.setdefault("title", "xyz")
print(x)

spiderman
```

```
In [51]: movie = { "genre": "comedy",
                  "title": "spiderman",
                  "year": 2000
                }
x = movie.setdefault("length", "2 hours")
print(x)
print(movie)
```

```
2 hours
{'genre': 'comedy', 'title': 'spiderman', 'year': 2000, 'length': '2 hours'}
```

```
In [55]: """ Python | Sort Python Dictionaries by Key """
p = {"apple":1, "tamrind":2, "orange":3, "sugarcane": 4}
a = list(p.keys())
print(a)
a.sort()
print(a)
sorted_dict = {i:p[i] for i in a}
print(sorted_dict)
```

```
['apple', 'tamrind', 'orange', 'sugarcane']
['apple', 'orange', 'sugarcane', 'tamrind']
{'apple': 1, 'orange': 3, 'sugarcane': 4, 'tamrind': 2}
```

```
In [56]: """ user input - dictionary """
a = {}
n = int(input("enter the number of items you want in the dictionary:"))
for i in range(n):
    b = input("Enter keys")
    c = input("enter values")
    a.update({b:c})
    print(a)
```

```
enter the number of items you want in the dictionary:4
Enter keys"apple"
enter values14
{'"apple"' : '14'}
Enter keysboy
enter values20
{'"apple"' : '14', 'boy': '20'}
Enter keys13
enter values"cherry"
{'"apple"' : '14', 'boy': '20', '13': '"cherry"' }
Enter keys18
enter values23.56
{'"apple"' : '14', 'boy': '20', '13': '"cherry"', '18': '23.56'}
```

```
In [59]: """ take the input from the user aand reverse the dictionary"""
a = {}
n = int(input("enter the number of items you want in the dictionary"))
for i in range(n):
    b = input("enter keys ")
    c = input("enter values")
    a.update({b:c})
print(a)
m = list(a.items())
print(m)
m.reverse()
print(m)
print(dict(m))
```

```
enter the number of items you want in the dictionary3
enter keys 14
enter valuespython
enter keys 15
enter valueshello
enter keys 16
enter valuescode
{'14': 'python', '15': 'hello', '16': 'code'}
[('14', 'python'), ('15', 'hello'), ('16', 'code')]
[('16', 'code'), ('15', 'hello'), ('14', 'python')]
{'16': 'code', '15': 'hello', '14': 'python'}
```

```
In [60]: """ find out max and min key and values in dictionary"""
a = { 12:"horse", 19:"lion", 25:"monkey", 2:"leopard"}
b = list(a.keys())
c = list(a.values())
print(max(b))
print(min(b))
print(max(c))
print(min(c))
```

```
25
2
monkey
horse
```

In []: